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PUBLIC SECTOR · DEFENSE INDUSTRIAL BASE

Elasticsearch, Inc.
4100 Fairfax Dr, 5th Floor
Arlington, VA 22203
www.elastic.co

TO: Air Force SBIR/STTR Contracting Office · AFWERX / DAF SBIR Program Office

FROM: Elasticsearch, Inc. (Elastic) · Defense Industrial Base & SLED Divisions

DATE: November 2026

SUBJECT: Commercial Transition Support & Technology Integration Alignment for Agile Defense Systems LLC

AFWERX SBIR Phase I Proposal: Cryptographic Execution Provenance for Autonomous Tactical Edge Enclaves

Dear AFWERX SBIR Selection Committee,

On behalf of Elasticsearch, Inc. ("Elastic"), we are pleased to provide this letter of commercial transition support for the proposal submitted by **Agile Defense Systems LLC**, titled *"Cryptographic Execution Provenance for Autonomous Tactical Edge Enclaves (osiris Runtime)."*

As an established provider of mission-critical search, observability, and security solutions across the U.S. Department of Defense, the Intelligence Community, and the Defense Industrial Base (DIB), Elastic recognizes the urgent necessity for active execution governance, post-quantum audit readiness, and zero-trust verification in autonomous computing environments.

Operational Relevance to Defense Missions

As the Department of the Air Force accelerates the deployment of autonomous systems, edge AI decision-support tools, and automated compiler toolchains into sensitive enclaves, establishing unalterable operational audit trails is critical for CMMC 2.0 Level 2/3, CNSA 2.0 Post-Quantum Cryptography, and NIST SP 800-171/53 compliance. Traditional plain-text logging mechanisms remain vulnerable to interception, manipulation, and unauthorized modification by elevated processes or insider adversaries. Furthermore, passive SIEM logging only captures damage after execution occurs.

Agile Defense Systems' **osiris** architecture addresses this challenge directly at the tactical edge by enforcing deterministic, payload-bound execution gates (SYS-021), hybrid post-quantum signatures (NIST FIPS 204 ML-DSA-65 + Ed25519), and Merkle hash-chained audit logging, ensuring that every mutating machine operation is cryptographically signed and non-repudiable before execution.

Technology Integration & Downstream Commercial Transition

Agile Defense Systems has designed its operational telemetry to conform directly with the **Elastic Common Schema (ECS 8.x)**, enabling immediate, high-throughput ingestion into Elasticsearch and Elastic Security (SIEM).

In support of this Phase I effort and prospective Phase II commercialization, Elastic intends to collaborate with Agile Defense Systems to:

1. **Validate ECS Telemetry Integration:** Evaluate the ingestion of osiris edge execution telemetry, post-quantum signatures, and Merkle ledger blocks into Elasticsearch clusters deployed in FedRAMP-authorized, air-gapped, and disconnected tactical defense configurations.
2. **Co-Develop Security Detection Content:** Explore out-of-the-box Elastic Security correlation rules for detecting unauthorized autonomous agent tool calls, in-flight payload hash divergences, and execution authority escalations.
3. **Assess Joint Customer Pilot Opportunities:** Identify prospective Defense Industrial Base suppliers, research enclaves, and academic partner testbeds (such as University of the Cumberlands) interested in replacing high-cost, volume-penalized legacy SIEM tools (such as Splunk) with a modern, verifiable Elastic-osiris zero-trust monitoring stack.
4. **Phase II Teaming Alignment:** Explore joint participation in an AFWERX SBIR Phase II prototype deployment ($1.8M) to integrate osiris cryptographic edge sensors with enterprise Elastic Security fabrics across Air Force operational environments.

We believe Agile Defense Systems' approach represents an innovative and practical leap forward in verifiable computing for defense applications. We strongly endorse their Phase I proposal and look forward to collaborating on its technical transition.

Sincerely,

Sarnath Chari
Enterprise Account Executive — Defense Industrial Base
Elasticsearch, Inc.
sarnath.chari@elastic.co

Christian Martucci
Account Executive — SLED Central
Elasticsearch, Inc.
christian.martucci@elastic.co